Theorem 1 (Lindemann–Weierstrass)**.** *Let $S \subset \overline{\mathbb{Q}}$ be a finite set. Then the numbers $\{e^{\alpha}\}_{\alpha \in S}$ are linearly independent over $\overline{\mathbb{Q}}$.*

As a corollary, we have that both π and e are transcendental

Corollary 2. *For any non-zero algebraic number $\alpha \in \overline{\mathbb{Q}}$, the number e^{α}, is transcendental. In particular π and e are transendental.*

Proof. Let $S = \{0, \alpha\}$. If $e^{\alpha} \in \overline{\mathbb{Q}}$, then $e^{\alpha} \cdot e^{0} - e^{\alpha} = 0$, contradicting that **Theorem 1,** i.e, $\{e^{0}, e^{\alpha}\}$ are linearly independent over $\overline{\mathbb{Q}}$.

Immediatly, since 1 is algebraic, the number e is transendental. Moreover, if π was algebraic, then the number $i\pi$ would be algebraic. Hence $e^{i\pi} = -1$ is transcendental, a contradiction. Thus π is transcendental.

We must now prove the following theorem, to help us prove the main result.

Theorem 3. *Let $S \subset \overline{\mathbb{Q}}$ such that for all $\alpha \in S$, the set S contains all of it's conjugates, i.e, all elements that share the same minimal polynomial over $\mathbb{Q}$. Let $\lambda_{\alpha} \in \mathbb{Q}$ and suppose that $\lambda_{\alpha} = \lambda_{\beta}$ whenever α and β are conjugates. Then if*

$$\sum_{\alpha \in S} \lambda_{\alpha} e^{\alpha} = 0,$$

then every coefficient is 0.

Proof. Remove all $\alpha \in S$ for which $\lambda_{\alpha} = 0$, and now for a contradiction suppose that $S \neq \varnothing$. Let $\Phi(x) \in \mathbb{Z}[x]$ be a polynomial having each element of S as a simple root. Indeed such polynomial exists. Since every element of S is algebraic over $\mathbb{Q}$, for every set of conjugates in S multiply their minimal polynomials togethor and scale the coeffficients so that their product is in $\mathbb{Z}[x]$.

Now let $d = \deg \Phi$ and c be the leading coefficient of Φ. Let $n \geq 1$ be an integer, fix $\alpha \in S$ and let

$$f_n(x) = \frac{\Phi(x)^n}{x - \alpha}.$$

First observe that $f_n(x) \in \overline{\mathbb{Q}}[x]$ since the minimal polynomial of α, $(x - \alpha)$ must divide any polynomial have α as root, in particular it divides $\Phi(x)^n$. Let $f_n^{(j)}$ denote the j-th derivative and define

$$F_n(t) = f_n(t) + f_n^{(1)}(t) + f_n^{(3)}(t) + \cdots$$

which is indeed a finite sum since $f_n(t) \in \overline{\mathbb{Q}}[t]$ is a polynomial. Moreover, since $-e^{-x}F_n(x)$ is a primitive function of $-e^{-x}f_n(x)$, by Hermite's Formula,

$$\int_0^1 e^{z(1-t)} f_n(zt) z dt = e^{z} \int_0^z e^{-t} f_n(t) dt = e^{z} F_n(0) - F_n(z)$$

for all $z \in \mathbb{C}$. Let $\beta \in S$, by hypothesism

$$\sum_{\beta \in S} \lambda_\beta e^\beta = 0.$$

It follows that

$$\sum_{\beta \in S} \lambda_\beta F_n(\beta) = -\sum_{\beta \in S} \lambda_\beta \int_0^1 e^{\beta(1-t)} f_n(\beta t) \beta dt.$$

Since $f_n(x)$ is a polynomial of degree $nd - 1$, the integrals are $\mathcal{O}(m^{nd})$ where m is a large enough positive real number that only depends on S and $\Phi(x)$. Thus

$$\sum_{\beta \in S} \lambda_\beta F_n(\beta) = \mathcal{O}(m^{nd}).$$

Now we make a scaling that will force the right-hand side to go to 0 as $n \to \infty$. Set

$$g_n(x) = \frac{c^n \Phi(x)^n}{n!}, \qquad f_n(x) = \frac{g_n(x)}{x - \alpha}.$$

Then $f_n(x) \in \overline{\mathbb{Q}}[x]$ and $\deg f_n = nd - 1$ as before. Define

$$F_n(t) = \sum_{k \geq 0} f_n^{(k)}(t),$$

which is a finite sum since f_n is a polynomial. Repeating the argument above (Hermite's formula) we again obtain

$$\Sigma_n := \sum_{\beta \in S} \lambda_\beta F_n(\beta) = -\sum_{\beta \in S} \lambda_\beta \int_0^1 e^{\beta(1-t)} f_n(\beta t)\, \beta \, dt.$$

Since the set S is finite, there exists a constant $A > 0$ (depending only on S) such that $|e^{\beta(1-t)}\beta| \leq A$ for all $\beta \in S$ and $t \in [0, 1]$. Moreover, since βt ranges over a compact set (again because S is finite and $t \in [0, 1]$), there exists $B > 0$ such that

$$|\Phi(\beta t)| \leq B \quad \text{for all } \beta \in S,\ t \in [0, 1].$$

Hence for all such β, t,

$$|g_n(\beta t)| = \left| \frac{c^n \Phi(\beta t)^n}{n!} \right| \leq \frac{|c|^n B^n}{n!}.$$

Since also $|(\beta t - \alpha)^{-1}|$ is bounded on $t \in [0, 1]$ for each fixed β (note $\beta t = \alpha$ would force $t = 1$ and $\beta = \alpha$, but f_n is a polynomial so the cancellation has already occurred), we conclude that there exists $C > 0$ and $M > 0$ depending only on S and Φ such that

$$\left| \int_0^1 e^{\beta(1-t)} f_n(\beta t)\, \beta \, dt \right| \leq C \frac{M^{nd}}{n!}.$$

Summing over the finitely many $\beta \in S$ gives

$$|\Sigma_n| \le C' \frac{M^{nd}}{n!} \quad \Longrightarrow \quad \Sigma_n \to 0 \text{ as } n \to \infty.$$

We now show that, after clearing denominators once, Σ_n is an integer for every n.

Claim 1. There exists a positive integer D (independent of n) such that for every $n \ge 1$,

$$D\Sigma_n \text{ is an algebraic integer.}$$

Proof of Claim 1. Fix $\beta \in S$. By construction $\Phi \in \mathbb{Z}[x]$ and c is its leading coefficient, hence $c\beta$ is an algebraic integer. Since $g_n(x) = c^n \Phi(x)^n/n!$ has coefficients in $\mathbb{Q}$, multiplying by a fixed integer clears all denominators coming from the λ_β and the finitely many algebraic numbers occurring in the coefficients of f_n and its derivatives. Thus there exists $D \ge 1$ such that for all $\beta \in S$ and all $k \ge 0$,

$$D\, f_n^{(k)}(\beta)$$

is an algebraic integer. Therefore $D\, F_n(\beta)$ is an algebraic integer for every $\beta \in S$, and so is the $\mathbb{Q}$-linear combination $D\Sigma_n = \sum_{\beta \in S} \lambda_\beta\, (DF_n(\beta))$. □

Claim 2. $D\Sigma_n \in \mathbb{Z}$ for every $n \ge 1$.

Proof of Claim 2. Let σ be any $\mathbb{Q}$-embedding of the field generated by S into $\mathbb{C}$. Since $\Phi \in \mathbb{Z}[x]$, we have $\sigma(\Phi) = \Phi$, and since S contains all conjugates, σ permutes S. Also, by hypothesis $\lambda_\alpha = \lambda_\beta$ whenever α, β are conjugate, hence

$$\sigma\left(\sum_{\beta \in S} \lambda_\beta F_n(\beta)\right) = \sum_{\beta \in S} \lambda_{\sigma(\beta)}\, \sigma(F_n(\beta)) = \sum_{\beta \in S} \lambda_\beta F_n(\beta) = \Sigma_n.$$

Thus Σ_n is fixed by every σ, hence $\Sigma_n \in \mathbb{Q}$. Therefore $D\Sigma_n \in \mathbb{Q}$ as well. Combining this with Claim 1, we have $D\Sigma_n$ is both a rational number and an algebraic integer, so $D\Sigma_n \in \mathbb{Z}$. □

Finally, we show that Σ_n is *not* identically 0. Indeed, consider the term corresponding to $\beta = \alpha$. Since $f_n(x) = g_n(x)/(x - \alpha)$ and $g_n(\alpha) = 0$ but α is a *simple* root of Φ, one checks that $f_n(\alpha) \ne 0$ (in fact it is proportional to $\Phi'(\alpha)^n/n!$), and hence $F_n(\alpha) \ne 0$. As we assumed $\lambda_\alpha \ne 0$, this implies that for infinitely many n we have $\Sigma_n \ne 0$.

Now combine everything: for infinitely many n,

$$0 \ne D\Sigma_n \in \mathbb{Z}, \qquad \text{but} \qquad |D\Sigma_n| \le D\, C' \frac{M^{nd}}{n!} \xrightarrow[n \to \infty]{} 0.$$

Thus for all sufficiently large n with $\Sigma_n \ne 0$ we get $0 < |D\Sigma_n| < 1$, contradicting that $D\Sigma_n$ is an integer.

This contradiction shows that our assumption $S \ne \varnothing$ was false. Therefore $S = \varnothing$, i.e.

every coefficient λ_α must be 0, as desired.

Theorem 1. *Let $S \subset \overline{\mathbb{Q}}$ be a finite set. Then the numbers $\{e^\alpha\}_{\alpha \in S}$ are linearly independent over $\overline{\mathbb{Q}}$.*

Proof. Suppose for a contradiction that $\{e^\alpha\}_{\alpha \in S}$ is not linearly independent over $\overline{\mathbb{Q}}$. Then there exist distinct $\alpha_1, \dots, \alpha_n \in S$ and coefficients $c_1, \dots, c_n \in \overline{\mathbb{Q}}$, not all 0, such that

$$\sum_{i=1}^{n} c_i e^{\alpha_i} = 0. \tag{1}$$

Choose such a relation with n minimal, so in particular each $c_i \neq 0$. Let $L/\mathbb{Q}$ be a finite Galois extension containing $\alpha_1, \dots, \alpha_n$ and $c_1, \dots, c_n$, and set $G = \mathrm{Gal}(L/\mathbb{Q})$. Fix some index j and note $c_j \neq 0$. Consider the $\mathbb{Q}$-linear functional

$$T_j : L \to \mathbb{Q}, \qquad T_j(x) = \mathrm{Tr}_{L/\mathbb{Q}}(x\, c_j).$$

This is not the zero map (equivalently, the trace pairing on L is nondegenerate), hence we may choose $a \in L$ such that $\mathrm{Tr}_{L/\mathbb{Q}}(ac_j) \neq 0$. Now apply $\sigma \in G$ to (1) and multiply by $\sigma(a)$:

$$0 = \sigma(a) \sum_{i=1}^{n} \sigma(c_i) e^{\sigma(\alpha_i)}.$$

Summing over all $\sigma \in G$ gives

$$0 = \sum_{\sigma \in G} \sum_{i=1}^{n} \sigma(ac_i)\, e^{\sigma(\alpha_i)}. \tag{2}$$

Let

$$T := \{\sigma(\alpha_i) : \sigma \in G,\ 1 \le i \le n\},$$

so T is finite and contains all conjugates of each of its elements. For $\beta \in T$, define

$$\lambda_\beta = \sum_{\substack{\sigma \in G,\ 1 \le i \le n \\ \sigma(\alpha_i) = \beta}} \sigma(ac_i).$$

Then (2) becomes

$$\sum_{\beta \in T} \lambda_\beta e^\beta = 0. \tag{3}$$

We now check the hypotheses of the previous theorem. First, for any $\tau \in G$ we have $\tau(T) = T$, and moreover

$$\tau(\lambda_\beta) = \sum_{\substack{\sigma, i \\ \sigma(\alpha_i) = \beta}} \tau(\sigma(ac_i)) = \sum_{\substack{\sigma, i \\ (\tau\sigma)(\alpha_i) = \tau(\beta)}} (\tau\sigma)(ac_i) = \lambda_{\tau(\beta)}.$$

In particular, if $\tau(\beta) = \beta$ then $\tau(\lambda_\beta) = \lambda_\beta$, so $\lambda_\beta \in L^G = \mathbb{Q}$. Also, if β, β' are conjugate then $\beta' = \tau(\beta)$ for some τ, and the above gives $\lambda_{\beta'} = \lambda_{\tau(\beta)} = \tau(\lambda_\beta) = \lambda_\beta$ (since $\lambda_\beta \in \mathbb{Q}$). Thus $\lambda_\beta \in \mathbb{Q}$ and the coefficients are constant on conjugacy classes.

Finally, (3) is not the zero relation. Indeed, consider $\beta = \alpha_j \in T$. The contribution to λ_{α_j} coming from the terms with $i = j$ is

$$\sum_{\substack{\sigma \in G \\ \sigma(\alpha_j) = \alpha_j}} \sigma(ac_j) = \mathrm{Tr}_{L^H/\mathbb{Q}}(ac_j) \quad \text{where } H = \mathrm{Stab}_G(\alpha_j),$$

and in particular the full sum over all $\sigma \in G$ contains $\mathrm{Tr}_{L/\mathbb{Q}}(ac_j) \neq 0$ among its orbit-sums, so not all λ_β can vanish. Concretely, since

$$\sum_{\sigma \in G} \sigma(ac_j) = \mathrm{Tr}_{L/\mathbb{Q}}(ac_j) \neq 0,$$

at least one coefficient λ_β coming from the j-orbit is nonzero.

Therefore we have produced a nontrivial relation (3) with T closed under conjugation and with $\lambda_\beta \in \mathbb{Q}$ constant on conjugacy classes. This contradicts **Theorem 4**, which forces every $\lambda_\beta = 0$. Hence our assumption was false, and $\{e^\alpha\}_{\alpha \in S}$ is linearly independent over $\overline{\mathbb{Q}}$.

References

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